S32K3 Real-Time Drivers AUTOSAR 4.9 Version 7.0.1

Version 7.0.1

Rev. 1 — 27 February 2026 **Release notes**

1 Getting Started

1.1 Package content

This release contains the NXP S32K3 Real-Time Drivers Version 7.0.1:

- "eclipse/plugins/<mod>_TS_T40D34M70I1R0" directories - Tresos Plugins, 1 per module.
- "SW32K3_S32M27x_RTD_R23-11_7.0.1_D2603.exe"
- "SW32K3_S32M27x_RTD_R23-11_7.0.1_D2603_ReleaseNotes.pdf" - This file.
- "SW32K3_S32M27x_RTD_R23-11_7.0.1_D2603_SBOM.spdx.json"
- "SW32K3_S32M27x_RTD_R23-11_7.0.1_D2603_DesignStudio_updatesite.zip"
- "SW32K3_S32M27x_RTD_R23-11_7.0.1_D2603_SafetyPackage.zip" - contains FMEA reports and Safety Manual.
- "SW32K3_S32M27x_RTD_R23-11_7.0.1_D2603_QualityPackage.zip"
- Various other files: GettingStarted.htm start page and associated images, the license.txt EULA file and the Uninstall.exe utility for removing the RTD installation.

1.2 Installation Design Studio

1.2.1 Bundled in S32 Design Studio

S32 RTD is delivered bundled in the S32 Design Studio. In this case, it's already configured and ready to use. New S32DS project wizard and New S32DS project from Example can be used to create S32 RTD based projects.

1.2.2 Delivered as an extension for S32 Design Studio

S32 RTD is delivered as an Update Site for S32 Design Studio "*S32 Design Studio 3.6.4*". In this case, it must be installed by opening Help -> S32 Design Studio Extensions and Updates -> Add Update Sites and selecting the archive file containing the S32 RTD software and then check the S32 RTD software package to be installed and continue the installation process. After it is installed, New S32DS project wizard and New S32DS project from Example can be used to create S32 RTD based projects.

1.3 Installation EB Tresos

Follow the installer steps. By default the installer will create a link between the installation target directory and a selected EB Tresos installation. If you choose not to create a link, you can later create one manually or you can copy all "<mod>_TS_T40D34M70I1R0" directories and .JAR files to the "<Tresos Install Path>\plugins" directory.

2 Release Specifics

The S32K3 Real-Time Drivers Version 7.0.1 is AUTOSAR R23-11 compliant. The AUTOSAR Configuration ARXML specification takes precedence over AUTOSAR SWS PDF Specifications if there are discrepancies.

The S32K3 Real-Time Drivers Version 7.0.1 can be used also in non-AUTOSAR environment, as a collection of peripheral drivers designed to simplify and accelerate application development on NXP microcontrollers.

2.1 Release Details

This is the NXP S32K3_S32M27x Real-Time Drivers Release Version 7.0.1, qualified as RFP release in terms of quality, supporting AUTOSAR R23-11 and non AUTOSAR, with functionalities and tests on S32K358(Rev 1.1), S32K396(cut 1.1), S32K311, S32K342, S32K312, S32K344, S32K388, S32K389 and S32M276 (cut 2.0).

This release contains a deviation from AUTOSAR recommended version check inside source files for more details.

This is qualified as RFP for all drivers.

RFP qualified drivers can be used in production.

This release was tested using:

- Silicon P32K358GHMJBS 1P14E CTAA2336A (Rev1.1).
- Silicon P32K396EHMJBS 1P40E QAA2331A (E5) (cut 1.1).
- Silicon P32K311HVS 0P98C CTZA2242B (E5).
- Silicon P32K344EHVMMS 1P55A CTSB2128B.
- Silicon P32K312NHVPBS 0P09C CTZM2132B.
- Silicon P32K342EHVPBE 0P97C CTZM2139A.
- Silicon P32K388HHMJBS 0P39J CTAH2432L (E5).
- Silicon P32K389HHMJGS 0P23N CTAC2533J (E5).
- Silicon P32M276CC MKHS 0P69K 20230904 (E5-B0) (Use for CANTRCV).
- Silicon P32M276LC MKHS 0P69K 20230904 (E5-B0) (Use for LINTRCV).
- Mini Module XS32K3X8CVB-Q289 PCB 53108 RevX3 SCH RevA1.
- Mini Module XS32K396-BGA-DC PCB 54614 RevX1 SCH RevA.
- Mini Module XS32K3X2CVB-Q100 PCB 48306 RevX1 SCH RevA.
- Mini Module XS32K3XXCVB-Q257 PCB 47020 RevA1 SCH RevX2.
- Mini Module XS32K3X2CVB-Q172 PCB 48307 RevX1 SCH RevA.
- Mother Board X-S32K2XX-MB PCB 31431 RevX3 SCH RevB.
- Mother Board X-S32K3XXX-MB PCB 47386 RevA SCH RevA.
- Mother Board X-S32X-MB version A PCB 54935 RevX1 SCH RevA.
- EVB S32K3X4EVB-Q257 47827 PCB RevX1 SCH RevA.
- EVB S32K3X4EVB-T172 PCB 53148 RevX5 SCH RevB.
- EVB S32K312EVB-Q172 PCB 50892 RevA SCH RevB.
- EVB S32K3X8EVB-Q289 PCB 54870 RevX2 SCH RevB2
- EVB S32K388EVB-Q289 PCB 88925 RevX1 SCH RevA
- CVB XS32M2XXCVB-Q064 PCB 53099 RevX1 SCH RevA.
- CVB XS32K388CVB-Q289 PCB RevA SCH RevX3.
- CVB XS32K389CVB-Q437
- EVB S32K358 HyperRAM internal validation board (for Mem_Exfls only).
- HSE frimware release: S32K344: hse_fw_s32k344 0.2.55.0 (RFP).
- HSE frimware release: S32K312: hse_fw_s32k312 0.2.55.0 (RFP).

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- HSE frimware release: S32K342: hse_fw_s32k342 0.2.55.0 (RFP).
- HSE frimware release: S32K358: hse_fw_s32k358_0.2.55.0 (RFP).
- HSE frimware release: S32K311: hse_fw_sS2k311_0.2.55.0 (RFP).
- HSE frimware release: S32K396: hsse_fw_s32k396_0.2.50.0 (RFP).
- HSE frimware release: S32M276: hse_fw_s32m276_0.2.55.0 (RFP).
- HSE frimware release: S32K388: hse_fw_s32k388_0.2.49.0 (RFP).
- HSE frimware release: S32K389: hse_fw_s32k389_0.2.71.0 (RFP).
- VDK Simulation Version: S32K388: 09_R5.1.1_S32K388_Release (Smoke test for all drivers except AE, Zipwire, CanTrcv, LinTrcv, Otp, Dpga, Gdu, Messaging and Mem_eep).
- VDK Simulation Version: S32K324: 01_R1.0.1_S32K324_Release (Smoke test for all drivers except AE, Zipwire, CanTrcv, LinTrcv, Otp, Dpga, Gdu, Messaging and Mem_eep).

In all source files, Software Version values are checked (major, minor, patch). AUTOSAR release or SWS versions are not checked during preprocessing/template generation.

The correct SWS versions are exported by each module.

The functions contained in the Dem, Det, EcuC, EcuM, Rte, OS and interface plugins are sample stub functions.

These functions should be replaced by the user developed code during integration.

The Resource module is needed to select the MCU derivative.

The derivatives supported can be found in the Resource module definition file, parameter 'ResourceSubderivative'.

The following limitations are present in this release:

Known limitation:

- Due to low SRAM memory space on S32K310, S32K311, S32K312, S32K322, S32K341, S32K342, S32M274 and S32M276 derivatives, users should select FLASH profile to create new project on S32 Design Studio.
- S32K388 DS and EBT, S32K358 DS, S32K396 EB and DS Hld, S32K344 EB examples are not supported for the I2S driver.
- S32K358: Reading from Hyperflash with QSPI may result in corrupted data.
- The software supports up to 256 partitions, with IDs ranging from 0 to 255. Any ID outside the range might cause issues with Partition ID checking feature on some drivers.

Known issue with IAR compiler, all RTD drivers:

- Warning regarding stack usage is thrown for reference implementations of core exceptions in startup files when drivers are compiled with IAR. These functions are provided as reference code and can be replace/ modified by the application.
- Usage of IAR compiler option "-enable_stack_usage " will issue warnings regarding uncalled functions (eg : interrupt handlers). This should be disregarded.
- IAR cannot analyze stack usage for function in .s file and the function with indirect call(function pointer).

Known issue with GCC compiler, all RTD drivers:

- Warning regarding enum size is thrown by the linker due to usage of "-fno-short-enums" option: "use of enum values across objects may fail". The drivers do not use any library enum types - no functional impact.

Known issue for PE Micro debug plugins in Design Studio:

- Debugging an application in RAM might fail if another application is present in target flash (debug communication lost during startup, MCU boots the application in flash). Workaround proposed: erase flash before loading an application in RAM; this can be done with the following command launched in the PE for example GDB Server console: "pegdbserver_console.exe -device=NXP_S32K3xx_S32K344

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-programmingtype=3 -runafterprogramming=0 -interface=USBMULTILINK -startserver -singlesession -quitafterprogramming -flashobjectfile=path_to_a_srec_or_elf_file".

The following limitations are present in this release, beside the ones presented in Drivers Manuals:

- During this SW product development, the achieved code coverage is between 83.66% min and 100% max, as it can be seen in the individual Code Coverage Summary reports. Nevertheless, the whole code went through a diligently inspection-based review.

RTD Header Files reference

Table 1. RTD Header Files reference

Derivatives	RTD Header files
S32K344	S32K3 Reference Manual Rev. 3, 10/2021
S32K324	S32K3 Reference Manual Rev. 3, 10/2021
S32K314	S32K3 Reference Manual Rev. 3, 10/2021
S32K312	S32K3 Reference Manual Rev. 4, 04/2022
S32K311	S32K3 Reference Manual Rev. 7, 05/2023
S32K341	S32K3xx Reference Manual Rev.5, 09/2022
S32K342	S32K3xx Reference Manual Rev.5, 09/2022
S32K322	S32K3xx Reference Manual Rev.5, 09/2022
S32K358	S32K3xx Reference Manual Rev. 8_Update, 11/2024
S32K356	S32K3 Reference Manual Rev. 11, 07/2025
S32K388	S32K3 Reference Manual Rev. 9, 08/2024
S32K389	S32K3xx Reference Manual Rev. 10, 04/2025
S32K39, S32K37, S32K36	S32K39,S32K37 and S32K36 Reference Manual Rev.10, 04/2025
S32M276	S32M27x Reference Manual, Rev.4, 09/2024

2.2 Used Documentation

This release was developed and tested based on the following documents:

Table 2. Reference Manuals

Document Title	Version and Date
S32K3xx Reference Manual	Rev. 11, 07/2025
S32K3xx Data Sheet	Rev. 12, 07/2025
S32M27x Reference Manual	Rev.6, 05/2025
S32M2xx Data Sheet	Rev. 8 — 02/2025
S32K39, S32K37 and S32K36 Reference Manual	Rev. 4, 11/2024
S32K39 and S32K37 Data Sheet	Rev.4 — 06/2024

Table 3. Implemented Errata

Document Title	Maskset	Date
S32K358 Mask Set Errata for Mask	1P14E	Rev. 3 - 21 January 2025

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Table 3. Implemented Errata*...continued*

Document Title	Maskset	Date
S32K311 Mask Set Errata	0P98C	Rev. 5 - 21 January 2025
S32K396 Mask Set Errata for Mask	0P40E	Rev. FEB2024, 2/2024
S32K396 Mask Set Errata for Mask	1P40E	Rev. MAR2024, 3/2024
S32K312 Mask Set Errata for Mask	0P09C	Rev. 8 - 22 January 2025
S32K342 Mask Set Errata for Mask	0P97C	Rev. 6 - 21 January 2025
S32K344 Mask Set Errata for Mask	0P55A/1P55A	Rev. 3 - 21 January 2025
S32K388 Mask Set Errata for Mask	0P39J/0P24N	Rev. 2 - 21 January 2025
S32M276 Mask Set Errata for Mask	P98C+P69K	Rev. 1, 8/2024
S32K389 Mask Set Errata for Mask	mask_id	Rev. 1.1 — 18 September 2024

2.3 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices from NXP:

- s32k310_hdqfp100
- s32k310_lqfp48
- s32k311_hdqfp100 / MWCT2015S_hdqfp100
- s32k311_lqfp48
- s32k312_hdqfp100 / MWCT2016S_hdqfp100
- s32k312_lqfp100
- s32k312_hdqfp172 / MWCT2016S_hdqfp172
- s32k314_lqfp100
- s32k314_hdqfp172
- s32k314_mapbga257
- s32k322_hdqfp100 / MWCT2D16S_hdqfp100
- s32k322_hdqfp172 / MWCT2D16S_hdqfp172
- s32k324_lqfp100
- s32k324_hdqfp172 / MWCT2D17S_hdqfp172
- s32k324_mapbga257
- s32k341_hdqfp100
- s32k341_hdqfp172
- s32k342_hdqfp100
- s32k342_hdqfp172
- s32k344_lqfp100
- s32k344_hdqfp172
- s32k344_mapbga257
- s32k394_mapbga289
- s32k394_lqfp176
- s32k396_mapbga289
- s32k396_lqfp176
- s32k374_mapbga289
- s32k374_lqfp176
- s32k376_mapbga289

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- s32k376_lqfp176
- s32k364_mapbga289
- s32k364_lqfp176
- s32k366_mapbga289
- s32k366_lqfp176
- s32k358_hdqfp172
- s32k358_mapbga289
- s32k356_hdqfp172
- s32k356_mapbga289
- s32k328_hdqfp172
- s32k328_mapbga289
- s32k338_hdqfp172
- s32k338_mapbga289
- s32k348_hdqfp172
- s32k348_mapbga289
- s32m274_lqfp64
- s32m276_lqfp64
- s32k388_mapbga289
- s32k389_mapbga437

Note: MWCT part numbers contain NXP confidential IP for Qi Wireless Power.

2.4 Modules Configuration

2.4.1 EB Tresos

Modules configurations were developed and tested using the Tresos Configuration Tool version "*EB Tresos Studio 30.0.0 b240617-0233*"

Configuration definition files were developed according to AUTOSAR 4.9.0, AUTOSAR_EcucParamDef.arxml

A folder named "<mod>_TS_TtDdMmIiRr" exists for each delivered module (<mod>). It is called a Tresos plugin for the module. A plugin contains the AUTOSAR module definition file (epd), the Tresos Xpath Data Model module definition file (xdm), the module user and integration manuals, the module configuration generation template source files, and the module driver static source files. Additional necessary Tresos specific tooling files are also included.

Plugin Encoding: <mod>_TS_TtDdMmIiRr

Important change related to the plugin notation:
- "m" = coding major and minor version number, can contain 1 or more digits
- "i" = patch number.
The major version number will be left out, if it is "0", in this case "m" contains 1 digit only,
otherwise it contains 2 digits

For this release:

- t=40, CortexM Architecture
- d=34, S32K3XX (derivative)
- m=70, Release major and minor version
- i=1, Release patch version
- r=0, Reserved

2.4.2 S32 Design Studio

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Configuration components were developed using "*S32 Design Studio 3.6.4*".

The components are split in three tools inside S32 Design Studio: Pins Tool, Clocks Tool, Peripherals tool which enable the generation of configuration structures to be used by both Autosar and low-level drivers.

2.5 Support and Driver Plugins Delivered

Table 4. Support and Driver Plugins Delivered

Plugin	Low level interface	SW Version	Description
ADC	Bctu_Ip, Adc_Sar_Ip, Sdadc_Ip	sw version 7.0.1	ASR, Analog to Digital Conversion
BASENXP	N/A	sw version 7.0.1	ASR Base Module, General AUTOSAR and Hardware Specific register files
CAN_43_ FLEXCAN	FlexCAN_ip	sw version 7.0.1	ASR, Controller Area Network
CRC	Crc_Ip	sw version 7.0.1	ASR, Cyclic Redundancy Check
DIO	Siul2_Dio	sw version 7.0.1	ASR, Digital Input Output
ETH_43_GMAC	GMAC_ip	sw version 7.0.1	ASR, Ethernet
FEE	N/A	sw version 7.0.1	ASR, Flash EEPROM Emulation
GPT	Emios_Gpt, Pit, Rtc, Stm	sw version 7.0.1	ASR, General Purpose Timer
I2C	FlexIO_I2C, LPI2C	sw version 7.0.1	ASR, Inter-Integrated Circuit
I2S	FlexIO_I2S, Sai_Ip	sw version 7.0.1	ASR, Inter-IC Sound
ICU	Emios_Icu, Cmp, Siul2_Icu, Wkpu	sw version 7.0.1	ASR, Input Capture Unit
LIN_43_ LPUART_ FLEXIO	FlexIo_Lin, LPUART	sw version 7.0.1	ASR, Local Interconnect Network
MCL	Cache_Ip, Dma_Ip, Lcu_Ip, Trgmux_Ip, Emios_Mcl	sw version 7.0.1	ASR, Microcontroller Library
MCU	Clock_Ip, Power_Ip, Ram_Ip, Siul2_Mcu, Emios_Mcu	sw version 7.0.1	ASR, Microcontroller Unit
OCU	Emios_Ocu	sw version 7.0.1	ASR, Output Control Unit
PLATFORM	startup, IntCtrl_Ip, Intm_Ip, Mpu_ M7_Ip, System_Ip	sw version 7.0.1	CDD, Platform
PORT	Siul2_Port, Tspc_Port	sw version 7.0.1	ASR, Port
PWM	Emios_Pwm, FlexIo_Pwm, Etpu_ Pwm	sw version 7.0.1	ASR, Pulse Width Modulation
RESOURCE	N/A	sw version 7.0.1	Resource Module, Required by all other modules to select MCU derivative
RM	Xbar_Ip, Xrdc_Ip, Sema42_ Ip, Xbic_Ip, Pflash_Ip, Virt_ Wrapper_Ip, IntCtrl_Ip, Dmamux_ Ip, Mscm_Ip	sw version 7.0.1	CDD, Resource Manager
SENT	FlexIo_Sent	sw version 7.0.1	ASR, Single edge nibble transmission
SPI	FlexIO_Spi, Lpspi, Dspi	sw version 7.0.1	ASR, Serial Peripheral Interface
UART	FlexIo_UART, LPUART	sw version 7.0.1	CDD, UART ASR

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Table 4. Support and Driver Plugins Delivered*...continued*

Plugin	Low level interface	SW Version	Description
ZIPWIRE	N/A	sw version 7.0.1	CDD, Zipwire
AE	Hvm_Ip, Aec_Ip	sw version 7.0.1	CDD, Aplication Extention
WDG	Swt_Ip, AeWdog_Ip	sw version 7.0.1	ASR, Watchdog
MEM_43_ INFLS	C40_Ip, Pflash_Ip	sw version 7.0.1	ASR, Mem Internal Flash
MEM_43_ EXFLS	Qspi_Ip	sw version 7.0.1	ASR, Mem External Flash
MEM_43_EEP	Sd_Emmc_Ip	sw version 7.0.1	ASR, Mem EEPROM
MEMACC	N/A	sw version 7.0.1	ASR, Memory access
MESSAGING	Mu_Ip	sw version 7.0.1	CDD, Messaging
GDU	Gdu_Ip	sw version 7.0.1	CDD, Gate ASR Unit
OCOTP	Mem_Otp_Ip	sw version 7.0.1	ASR, On-Chip One Time Programmable
LINTRCV_43_ AE	N/A	sw version 7.0.1	ASR, Lin Transceiver
CANTRCV_43_ AE	N/A	sw version 7.0.1	ASR, Can Transceiver
DPGA	Dpga_Ip	sw version 7.0.1	CDD, DPGA ASR
CANIF	N/A	sw version 7.0.1	Support Stub, Controller Area Network Interface
DEM	N/A	sw version 7.0.1	Support Stub, Diagnostic Event Manager
DET	N/A	sw version 7.0.1	Support Stub, Development Error Tracer
ECUC	N/A	sw version 7.0.1	Support Stub, ECU Configuration
ECUM	N/A	sw version 7.0.1	Support Stub, ECU State Manager
ETHIF	N/A	sw version 7.0.1	Support Stub, Ethernet Interface
ETHTRCV	N/A	sw version 7.0.1	Support stub, Ethernet Transceiver
ETHSWT	N/A	sw version 7.0.1	Support stub, Ethernet Switch
LINIF	N/A	sw version 7.0.1	Support Stub, Local Interconnect Network Interface
MEMIF	N/A	sw version 7.0.1	Support Stub, Memory Interface
OS	N/A	sw version 7.0.1	Support stub, Operating System
RTE	N/A	sw version 7.0.1	Support Stub, only for Schedule Manager
WDGIF	N/A	sw version 7.0.1	Support Stub, Watchdog Interface
MKA	N/A	sw version 7.0.1	Support Stub, Ethernet Interface

2.6 Module Plugin Folder Structure

Table 5. Module Plugin Folder Structure

Folder or file	Description
<mod>_TS_TtDdMmIiRr\anchors.xml	Tresos Configuration tooling documentation data file
<mod>_TS_TtDdMmIiRr\plugin.xml	Tresos Configuration tooling data file

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Table 5. Module Plugin Folder Structure*...continued*

Folder or file	Description
<mod>_TS_TtDdMmIiRr\ant_generator.xml	Tresos Configuration tooling data file
<mod>_TS_TtDdMmIiRr\autosar\<mod>.epd	Module Parameter Definition in AUTOSAR format
<mod>_TS_TtDdMmIiRr\config\<mod>.xdm	Module Parameter Definition in Tresos XDM format
<mod>_TS_TtDdMmIiRr\config_ext\<mod>PreConfiguration.xdm	Module Parameter Default Configuration in Tresos XDM format[1]
<mod>_TS_TtDdMmIiRr\doc\AUTOSAR_RTD_<mod>_IM.pdf	Module Integration Manual
<mod>_TS_TtDdMmIiRr\doc\AUTOSAR_RTD_<mod>_UM.pdf	Module User's Manual
<mod>_TS_TtDdMmIiRr\generate_PB	Post-build source files (only if applicable)
<mod>_TS_TtDdMmIiRr\generate_PB\src	Post-build source file templates (only if applicable)
<mod>_TS_TtDdMmIiRr\generate_PB\include	Post-build source file header templates (only if applicable)
<mod>_TS_TtDdMmIiRr\generate_LT	Link-time source files (only if applicable)
<mod>_TS_TtDdMmIiRr\generate_LT\src	Link-time source file templates (only if applicable)
<mod>_TS_TtDdMmIiRr\generate_LT\include	Link-time source file templates (only if applicable)
<mod>_TS_TtDdMmIiRr\generate_PC\	Pre-compile source files
<mod>_TS_TtDdMmIiRr\generate_PC\src	Pre-compile source files templates
<mod>_TS_TtDdMmIiRr\generate_PC\include	Pre-compile source files templates
<mod>_TS_TtDdMmIiRr\generate_swcd	Module BSWMD file
<mod>_TS_TtDdMmIiRr\include\	Module driver header files
<mod>_TS_TtDdMmIiRr\META-INF	Tresos Configuration tooling data and signature files
<mod>_TS_TtDdMmIiRr\src\	Module driver source files[2]

Notes:

[1] Not available for all plugins.

[2] The Support Stub Resource contains the "resource" folder instead of the "src" folder.

2.7 Compiler Options

This release was developed and tested with:

- NXP GCC 10.2.0 20200723
- Green Hills Multi 7.1.6d / Compiler 2021.1.4
- IAR ANSI C/C++ Compiler v.8.50.10 (safety version)
- Wind River Diab Compiler v7.0.6

2.7.1 DIAB Compiler/Linker/Assembler Options

Table 6. Compiler Options

Option	Description
-tARMCORTEXM7MG:simple	Selects target processor (hardware single-precision, software double-precision floating-point)

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Table 6. Compiler Options*...continued*

Option	Description
-mthumb	Selects generating code that executes in Thumb state
-std=c99	Follows the C99 standard for C
-Oz	Like -O2 with further optimizations to reduce code size
-g	Generates DWARF 4.0 debug information
-fstandalone-debug	Emits full debug info for all types used by the program
-Wstrict-prototypes	Warn if a function is declared or defined without specifying the argument types
-Wsign-compare	Produce warnings when comparing signed type with unsigned type
Wdouble-promotion	Give a warning when a value of type float is implicitly promoted to double
-Wunknown-pragmas	Issues a warning for unknown pragmas
-Wundef	Warns if an undefined identifier is evaluated in an #if directive. Such identifiers are replaced with zero
-Wextra	Enables some extra warning flags that are not enabled by '-Wall'
-Wall	Enables all of the most useful warnings (for historical reasons this option does not literally enable all warnings)
-pedantic	Emits a warning whenever the standard specified by the -std option requires a diagnostic
-Werror=implicit-function-declaration	Generates an error whenever a function is used before being declared
-fno-common	Compile common globals like normal definitions
-fno-signed-char	Char is unsigned
-fno-trigraphs	Do not process trigraph sequences
-V	Displays the current version number of the tool suite
-c	Stop after assembly and produce an object file for each source file
-DUSE_SW_VECTOR_MODE	Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode.
-DD_CACHE_ENABLE	Predefine D_CACHE_ENABLE as a macro, with definition 1. Enables data cache initalization in source file system.c under the Platform driver
-DI_CACHE_ENABLE	Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initalization in source file system.c under the Platform driver
-DMPU_ENABLE	Predefine MPU_ENABLE as a macro, with definition 1. Enable MPU default configuration from startup code under the Platform driver
-DENABLE_FPU	Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initalization in source file system.c under the Platform driver

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Table 6. Compiler Options*...continued*

Option	Description
-DMCAL_ENABLE_USER_MODE_SUPPORT	Predefine MCAL_ENABLE_USER_MODE_SUPPORT as a macro, with definition 1. Allows drivers to be configured in user mode.

Table 7. Assembler Options

Option	Description
-mthumb	Selects generating code that executes in Thumb state
-Xpreprocess-assembly	Invokes C preprocessor on assembly files before running the assembler
-Xassembly-listing	Produces an .lst assembly listing file
-c	Stop after assembly and produce an object file for each source file
-tARMCORTEXM7MG:simple	Selects target processor (hardware single-precision, software double-precision floating-point)

Table 8. Linker Options

Option	Description
-e Reset_Handler	Make the symbol Reset_Handler be treated as a root symbol and the start label of the application
linker_script_file.dld	Use linker_script_file.dld as the linker script. This script replaces the default linker script (rather than adding to it)
-m30	m2 + m4 + m8 + m16
-Xstack-usage	Gathers and display stack usage at link time
-Xpreprocess-lecl	Perform pre-processing on linker scripts
-Llibrary_path	Points to the libraries location for ARMV7EMMG to be used for linking
-lc	Links with the standard C library
-lm	Links with the math library
-tARMCORTEXM7MG:simple	Selects target processor (hardware single-precision, software double-precision floating-point)

2.7.2 GCC Compiler/Linker/Assembler Options

Table 9. Compiler Options

Option	Description
-mcpu=cortex-m7	Targeted ARM processor for which GCC should tune the performance of the code
-mthumb	Generates code that executes in Thumb state
-mlittle-endian	Generate code for a processor running in little-endian mode
-mfpu=fpv5-sp-d16	Specifies the floating-point hardware available on the target

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Table 9. Compiler Options*...continued*

Option	Description
-mfloat-abi=hard	Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions
-std=c99	Specifies the ISO C99 base standard
-Os	Optimize for size. Enables all -O2 optimizations except those that often increase code size
-ggdb3	Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program
-Wall	Enables all the warnings about constructions that some users consider questionable, and that are easy to avoid (or modify to prevent the warning), even in conjunction with macros
-Wextra	This enables some extra warning flags that are not enabled by -Wall
-pedantic	Issue all the warnings demanded by strict ISO C. Reject all programs that use forbidden extensions. Follows the version of the ISO C standard specified by the aforementioend -std option.
-Wstrict-prototypes	Warn if a function is declared or defined without specifying the argument types
-Wundef	Warn if an undefined identifier is evaluated in an #if directive. Such identifiers are replaced with zero
-Wunused	Warn whenever a function, variable, label, value, macro is unused
-Werror=implicit-function-declaration	Make the specified warning into an error. This option throws an error when a function is used before being declared
-Wsign-compare	Warn when a comparison between signed and unsigned values could produce an incorrect result when the signed value is converted to unsigned.
-Wdouble-promotion	Give a warning when a value of type float is implicitly promoted to double
-fno-short-enums	Specifies that the size of an enumeration type is at least 32 bits regardless of the size of the enumerator values.
-funsigned-char	Let the type char be unsigned by default, when the declaration does not use either signed or unsigned
-funsigned-bitfields	Let a bit-field be unsigned by default, when the declaration does not use either signed or unsigned
-fomit-frame-pointer	Omit the frame pointer in functions that don’t need one. This avoids the instructions to save, set up and restore the frame pointer; on many targets it also makes an extra register available.
-fno-common	Makes the compiler place uninitialized global variables in the BSS section of the object file. This inhibits the merging of tentative definitions by the linker so you get a multiple-

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Table 9. Compiler Options...*continued*

Option	Description
	definition error if the same variable is accidentally defined in more than one compilation unit
-fstack-usage	Makes the compiler output stack usage information for the program, on a per-function basis
-fdump-ipa-all	Enables all inter-procedural analysis dumps
-c	Stop after assembly and produce an object file for each source file
-DUSE_SW_VECTOR_MODE	Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode.
-DD_CACHE_ENABLE	Predefine D_CACHE_ENABLE as a macro, with definition 1. Enables data cache initalization in source file system.c under the Platform driver
-DI_CACHE_ENABLE	Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initalization in source file system.c under the Platform driver
-DMPU_ENABLE	Predefine MPU_ENABLE as a macro, with definition 1. Enable MPU default configuration from startup code under the Platform driver
-DENABLE_FPU	Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initalization in source file system.c under the Platform driver
-DMCAL_ENABLE_USER_MODE_SUPPORT	Predefine MCAL_ENABLE_USER_MODE_SUPPORT as a macro, with definition 1. Allows drivers to be configured in user mode.
--sysroot= $ (NEWLIB_DIR)	Specifies the path to the sysroot, for Cortex-M7 it is $ (TOOLCHAIN_DIR)/arm-none-eabi/newlib
-specs=nano.specs	Use Newlib nano specs
-specs=nosys.specs	Do not use printf/scanf

Table 10. Assembler Options

Option	Description
-Xassembler-with-cpp	Specifies the language for the following input files (rather than letting the compiler choose a default based on the file name suffix)
-mcpu=cortex-m7	Targeted ARM processor for which GCC should tune the performance of the code
-mfpu=fpv5-sp-d16	Specifies the floating-point hardware available on the target
-mfloat-abi=hard	Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions
-mthumb	Generates code that executes in Thumb state
-c	Stop after assembly and produce an object file for each source file

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Table 11. Linker Options

Option	Description
-Wl,-Map,filename	Produces a map filer
-T linkerfile	Use linkerfile as the linker script. This script replaces the default linker script (rather than adding to it)
--entry=Reset_Handler	Specifies that the program entry point is Reset_Handler
-nostartfiles	Do not use the standard system startup files when linking
-mcpu=cortex-m7	Targeted ARM processor for which GCC should tune the performance of the code
-mthumb	Generates code that executes in Thumb state
-mfpu=fpv5-sp-d16	Specifies the floating-point hardware available on the target
-mfloat-abi=hard	Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions
-mlittle-endian	Generate code for a processor running in little-endian mode
-ggdb3	Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program
-lc	Link with the C library
-lm	Link with the Math library
-lgcc	Link with the GCC library
-specs=nano.specs	Use Newlib nano specs
-specs=nosys.specs	Do not use printf/scanf
--sysroot=$ (LIB_DIR)	Specifies the path to the sysroot, for Cortex-M7, it is $ (TOOLCHAIN_DIR)/arm-none-eabi/newlib

2.7.3 GHS Compiler/Linker/Assembler Options

Table 12. Compiler Options

Option	Description
-cpu=cortexm7	Selects target processor: Arm Cortex M7
-thumb	Selects generating code that executes in Thumb state
-fpu=vfpv5_d16	Specifies hardware floating-point using the v5 version of the VFP instruction set, with 16 double-precision floating-point registers
-fsingle	Use hardware single-precision, software double-precision FP instructions
-c99	Specifies the ISO C99 base standard
--ghstd=last	Use the most recent version of Green Hills Standard mode (which enables warnings and errors that enforce a stricter coding standard than regular C and C++)
-Osize	Optimize for size
--gnu_asm	Enables GNU extended asm syntax support

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Table 12. Compiler Options*...continued*

Option	Description
-dual_debug	Generate DWARF 2.0 debug information
-G	Generate debug information
-keeptempfiles	Prevents the deletion of temporary files after they are used. If an assembly language file is created by the compiler, this option will place it in the current directory instead of the temporary directory
-Wimplicit-int	Produce warnings if functions are assumed to return int
-Wshadow	Produce warnings if variables are shadowed
-Wtrigraphs	Produce warnings if trigraphs are detected
-Wundef	Produce a warning if undefined identifiers are used in #if preprocessor statements
--unsigned_chars	Let the type char be unsigned, like unsigned char
--unsigned_fields	Bitfelds declared with an integer type are unsigned
--no_commons	Allocates uninitialized global variables to a section and initializes them to zero at program startup
--no_exceptions	Disables C++ support for exception handling
--no_slash_comment	C++ style // comments are not accepted andgenerate errors
--prototype_errors	Controls the treatment of functions referenced or called when no prototype has been provided
--incorrect_pragma_warnings	Controls the treatment of valid #pragma directives that use the wrong syntax
-c	Stop after assembly and produce an object file for each source file
-DUSE_SW_VECTOR_MODE	Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode.
-DD_CACHE_ENABLE	Predefine D_CACHE_ENABLE as a macro, with definition 1. Enables data cache initalization in source file system.c under the Platform driver
-DI_CACHE_ENABLE	Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initalization in source file system.c under the Platform driver
-DMPU_ENABLE	Predefine MPU_ENABLE as a macro, with definition 1. Enable MPU default configuration from startup code under the Platform driver
-DENABLE_FPU	Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initalization in source file system.c under the Platform driver
-DMCAL_ENABLE_USER_MODE_SUPPORT	Predefine MCAL_ENABLE_USER_MODE_SUPPORT as a macro, with definition 1. Allows drivers to be configured in user mode.

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Table 13. Assembler Options

Option	Description
-cpu=cortexm7	Selects target processor: Arm Cortex M7
-fpu=vfpv5_d16	Specifies hardware floating-point using the v5 version of the VFP instruction set, with 16 double-precision floating-point registers
-fsingle	Use hardware single-precision, software double-precision FP instructions
-preprocess_assembly_files	Controls whether assembly files with standard extensions such as .s and .asm are preprocessed
-list	Creates a listing by using the name and directory of the object file with the .lst extension
-c	Stop after assembly and produce an object file for each source file

Table 14. Linker Options

Option	Description
-T linker_script_file.ld	Use linker_script_file.ld as the linker script. This script replaces the default linker script (rather than adding to it)
-e Reset_Handler	Make the symbol Reset_Handler be treated as a root symbol and the start label of the application
-map	Produce a map file
-keepmap	Controls the retention of the map file in the event of a link error
-Mn	Generates a listing of symbols sorted alphabetically/ numerically by address
-delete	Instructs the linker to remove functions that are not referenced in the final executable. The linker iterates to find functions that do not have relocations pointing to them and eliminates them
-ignore_debug_references	Ignores relocations from DWARF debug sections when using -delete. DWARF debug information will contain references to deleted functions that may break some third-party debuggers
-Llibrary_path	Points to library_path (the libraries location) for thumb2 to be used for linking
-larch	Link architecture specific library
-lstartup	Link run-time environment startup routines. The source code for themodules in this library is provided in the src/libstartup directory
-lind_sd	Link language-independent library, containing support routines for features such as software floating point, run-time error checking, C99 complex numbers, and some general purpose routines of the ANSI C library

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Table 14. Linker Options*...continued*

Option	Description
-v	Prints verbose information about the activities of the linker, including the libraries it searches to resolve undefined symbols
-keep=C40_Ip_AccessCode	Avoid linker remove function C40_Ip_AccessCode from Fls module because it is not referenced explicitly
-nostartfiles	Controls the start files to be linked into the executable

2.7.4 IAR Compiler/Linker/Assembler Options

Table 15. Compiler Options

Option	Description
--cpu=Cortex-M7	Targeted ARM processor for which IAR should tune the performance of the code
--cpu_mode=thumb	Generates code that executes in Thumb state
--endian=little	Generate code for a processor running in little-endian mode
--fpu=FPv5-SP	Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). Single-precision variant.
-e	Enables all IAR C language extensions
-Ohz	Optimize for size. the compiler will emit AEABI attributes indicating the requested optimization goal. This information can be used by the linker to select smaller or faster variants of DLIB library functions
--debug	Makes the compiler include debugging information in the object modules. Including debug information will make the object files larger
--no_clustering	Disables static clustering optimizations. Static and global variables defined within the same module will not be arranged so that variables that are accessed in the same function are close to each other
--no_mem_idioms	Makes the compiler not optimize certain memory access patterns
--do_explicit_zero_opt_in_named_sections	Disable the exception for variables in user-named sections, and thus treat explicit initializations to zero as zero initializations, not copy initializations
--require_prototypes	Force the compiler to verify that all functions have proper prototypes. Generates an error otherwise
--no_wrap_diagnostics	Does not wrap long lines in diagnostic messages
--diag_suppress=Pa050	Suppresses diagnostic message Pa050
-DIAR	Predefine IAR as a macro, with definition 1
-DUSE_SW_VECTOR_MODE	Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode.

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Table 15. Compiler Options*...continued*

Option	Description
-DD_CACHE_ENABLE	Predefine D_CACHE_ENABLE as a macro, with definition 1. Enables data cache initalization in source file system.c under the Platform driver
-DI_CACHE_ENABLE	Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initalization in source file system.c under the Platform driver
-DMPU_ENABLE	Predefine MPU_ENABLE as a macro, with definition 1. Enable MPU default configuration from startup code under the Platform driver
-DENABLE_FPU	Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initalization in source file system.c under the Platform driver
-DMCAL_ENABLE_USER_MODE_SUPPORT	Predefine MCAL_ENABLE_USER_MODE_SUPPORT as a macro, with definition 1. Allows drivers to be configured in user mode.

Table 16. Assembler Options

Option	Description
--cpu Cortex-M7	Targeted ARM processor for which IAR should generate the instruction set
--fpu VFPv5-SP	Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). Single-precision variant.
--cpu_mode thumb	Selects the thumb mode for the assembler directive CODE
-g	Disables the automatic search for system include files
-r	Generates debug information

Table 17. Linker Options

Option	Description
--map filename	Produces a map file
--config linkerfile	Use linkerfile as the linker script. This script replaces the default linker script (rather than adding to it)
--cpu=Cortex-M7	Selects the ARM processor variant to link the application for
--fpu=FPv5-SP	Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). Single-precision variant.
--entry _start	Treats _start as a root symbol and start label
--enable_stack_usage	Enables stack usage analysis. If a linker map file is produced, a stack usage chapter is included in the map file
--skip_dynamic_initialization	Dynamic initialization (typically initialization of C++ objects with static storage duration) will not be performed automatically during application startup

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Table 17. Linker Options...*continued*

Option	Description
--no_wrap_diagnostics	Does not wrap long lines in diagnostic messages

2.8 Examples and Demos

Driver examples are provided and included in each plugin.

3 Known Issues for S32K3 RTD 7.0.1

3.1 Known Issues for 7.0.1

Issue key	Summary	Notes for Integrator	Severity
ARTD-188302 base	[BASE] Incorrect Prototype for Os If_Software_Semaphore_Lock function	Updated the function prototype to: boolean OsIf_Software_Semaphore_Lock(volatile uint32 * const Semaphore, uint32 Lockval)	3 - Moderate (apparent functional problem)
ARTD-165081 mcl	[Mcl] Fix misra violation rule 8.6 finding from safety team		3 - Moderate (apparent functional problem)
ARTD-165124 platform	[Platform] Fix misra violation finding from safety team		3 - Moderate (apparent functional problem)
ARTD-165127 platform	[Platform] MISRA 5.9 aIntmChannel Config is already used to represent an object with internal linkage.		3 - Moderate (apparent functional problem)
ARTD-163682 mcl	[MCL] Fix misra rule 2.8 and compiler warning report has some warnings		3 - Moderate (apparent functional problem)
ARTD-9734 can	[can] Errors and missing exclusive areas in Bswmd.arxml		3 - Moderate (apparent functional problem)
ARTD-10788 mcu	[MCU] IPs functions should not be visible at HLD layers		3 - Moderate (apparent functional problem)
ARTD-8477 mcu	[MCU] Check whether DFS divider is enabled in SetDfsMfiMfn function is not true		3 - Moderate (apparent functional problem)
ARTD-63297 mem_eep	[MEM_EEP] Error on EPD files, generate fail for IP layer		3 - Moderate (apparent functional problem)
ARTD-66280 LinTrcv	[LINTRCV] Bswmd file contains dummy implementation		3 - Moderate (apparent functional problem)
ARTD-142125 spi	[spi] Generated headers cannot be built in case sensitive file systems		3 - Moderate (apparent functional problem)

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Issue key	Summary	Notes for Integrator	Severity
ARTD-147656 spi	[SPI] Multiple OSApplications pointing to the same Partition generates build error		3 - Moderate (apparent functional problem)
ARTD-198226 mcu	[MCU]Lacks PRAMC2 support	Root Cause Details: Lacks PRAMC2 support -> come from the change between RM Proposed Solution: Update support PRAMC2	3 - Moderate (apparent functional problem)
ARTD-96858 mcu	[MCU][CMU] Improve the speed of the Clock_Ip_SetCmuFcFceRefCnt LfrefHfref	impact/effect: not impact to current implement.just improve the condition if else of function workaround: NA	3 - Moderate (apparent functional problem)
ARTD-60895 memacc	[MemAcc] - Missing #define memacc_config when Variant_no=1 and precompile		3 - Moderate (apparent functional problem)
ARTD-145180 mcu	[mcu] Generated headers cannot be built in case sensitive file systems		3 - Moderate (apparent functional problem)
ARTD-201180 lin	[TRAINS][LIN][DMA]: Different name of macro DMA channel configuration	impact/effect: The mismatch between the tags used for LIN and MCL leads to inconsistency in macro definitions, causing the build system to fail validation or generate incorrect code workaround: Ensure that the tags used for LIN and MCL are consistent and aligned with the expected macro definitions	3 - Moderate (apparent functional problem)
ARTD-143126 i2s	[S32K3XX_S32M27x_5.0.0][I2S]: Flexio DMA transmits incorrect data when BytesWidth is 8	Developer fixed by checking end of frame condition.	3 - Moderate (apparent functional problem)
ARTD-126874 mem_exfls	[Mem_ExFls] "Return value" field in ReqExport for Mem_Suspend and Mem_Resume is not correct - follow up	Follow requirement SWS_Mem_10024 and SWS_Mem_10025 Mem_Resume Mem_Suspend should return "void". But follow requirement: SWS_Mem_00083 and SWS_Mem_00084: those function should return Std_Return Type. They are inconsistent. Should be updated in the next Autosar version. Currently, the driver still return Std_ReturnType.	3 - Moderate (apparent functional problem)
ARTD-203019 pwm	[pwm] eMIOS duty switch between 100% and 0%	For OPWMCB_TRAIL_EDGE mode, add a new condition to force a Match B event when setting the duty cycle to 0% in Emios_Pwm_Ip_SetDutyCycleOpwmcb(), to fix the issue where PWM sometimes fails when switching the duty cycle from 100% to 0%. For OPWMCB_TRAIL_EDGE mode, implement a forced Match A event when setting the duty cycle to 100% in Emios_Pwm_Ip_SetDutyCycleOpwmcb(), to fix the issue where PWM	3 - Moderate (apparent functional problem)

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Issue key	Summary	Notes for Integrator	Severity
		sometimes fails when switching the duty cycle from 0% to 100%. For OPWMCB_LEAD_EDGE mode, implement a mechanism to update deadtime in Emios_Pwm_Ip_Set DutyCycleOpwmcb() to fix the issue where PWM sometimes fails when switching the duty cycle from 100% to 0%.	
ARTD-67356 mem_eep	[MEM_EEP] Wrong VendorApiInfix in .xdm		3 - Moderate (apparent functional problem)
ARTD-65880 memacc	[MemAcc] MemAccBufferAlignment Value can be not configed = 0	impact/effect: If MemAccBuffer AlignmentValue configure = 0 - > fail on Fee driver. workaround: Configure MemAccBufferAlignment Value > 0.	3 - Moderate (apparent functional problem)
ARTD-62541 memacc	[MemAcc]Errors in epd files	impact/effect: Build fail with EPD configuration workaround: No way to workaround	3 - Moderate (apparent functional problem)
ARTD-112604 fee	[Fee] Example config has incorrect ASR version	' impact/effect: Updated example config with the correct version so no impact workaround: Can not workaround	3 - Moderate (apparent functional problem)
ARTD-58529 memacc	[MemAcc] Services still return OK after retry reached 0	impact/effect: After retry is reach 0, the job is not complete but the driver still return the job result is OK workaround: Can not workaround	3 - Moderate (apparent functional problem)
ARTD-66563 memacc	[MemAcc] Implement the requirement SWS_MemAcc_00089 in the MemAcc_ActivateMem API (follow-up)	impact/effect: Mem driver binary is not check the format before using. It may lead to use wrong Mem driver binary. workaround: Nothing workaround	3 - Moderate (apparent functional problem)
ARTD-151686 spi	[Spi] ecvd missing some node relative of SpiBaudrateConfig		3 - Moderate (apparent functional problem)
ARTD-22338 ALL_GENERAL	[HF] [S32K3] RTD 1.0.0 Update Site - redundant ECPD files in Code Preview		3 - Moderate (apparent functional problem)
ARTD-19019 port	[S32R41 EAR] Port driver have to add some Det_reportError function following requirement		3 - Moderate (apparent functional problem)
ARTD-62554 mem_eep	[Mem_Eep] Build fail when disable MemDevErrorDetect node on EB	BlankcheckAPIs node should be kept disabled to avoid a build error.	3 - Moderate (apparent functional problem)
ARTD-203846 can	CAN Enhanced FIFO lost frame workaround.	impact/effect: <Add one sentence explaining how customer is impacted> workaround: <Add one sentence describing potential workaround>	3 - Moderate (apparent functional problem)

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Issue key	Summary	Notes for Integrator	Severity
ARTD-70430 mem_eep	[Mem_43_Eep]Different output when export/import between EB and DS	impact/effect: It might be that EPD import is incorrect. workaround: None	3 - Moderate (apparent functional problem)
ARTD-66034 memacc	[SAF85_S32R41 1.0.0][MemAcc] Data retrieved by MemAcc_Get MemoryInfo is incorrect when no job is running	impact/effect: Updated for the MemAcc_GetMemoryInfo API, retrieve memory information using "MemAcc_pConfigPtr" instead of MemAcc_pJobRuntimeInfo". workaround: No workaround	3 - Moderate (apparent functional problem)
ARTD-66122 memacc	[SAF85_S32R41 1.0.0][MemAcc] MemAcc_GetJobResult return MEMACC_MEM_OK while the Mems driver erase/write works failed	impact/effect: In the function Mem Acc_JobRequestMemService, make sure to return the result variable 'JobInfo->JobResult = MEMACC_MEM_FAILED' if the request is not accepted. workaround: No workaround	3 - Moderate (apparent functional problem)
ARTD-180645 psi5	[psi5] Implement check on partition assigned for correctness or out of bounds accesses	impact/effect: Driver code has been modified to support checks on assigned partitions.	3 - Moderate (apparent functional problem)
ARTD-17075 sent	[SENT] Incorrect name of variable types as coding guideline	Fixed in [ARTD-16949]	3 - Moderate (apparent functional problem)
ARTD-24293 uart	[UART] Error when import Uart HLD in S32DS	Implemented by ARTD-21727	3 - Moderate (apparent functional problem)
ARTD-112671 pwm	[Etpu] Minor issues in epd files of SW32K3_RTD_R21-11_4.0.0_P05	Impact/Effect: The customer is unable to generate or validate configuration due to a mismatch between the parameter's variant and the module's supported configuration variants, potentially blocking integration or build processes. Workaround: Update the parameter to use a configuration variant supported by the module (e.g., VARIANT_ PRE_COMPILE) or modify the module's supported variants to include VARIANT_POST_BUILD.	3 - Moderate (apparent functional problem)
ARTD-115962 pwm	[pwm] Errors of missing Etpu resources when generate example of PWM	Impact/Effect: Example generation fails for derivatives other than S32K396 due to lack of resources for etpu ip. Workaround: Check and use only S32K396 example.	3 - Moderate (apparent functional problem)
ARTD-114595 pwm	[ETPU] Build fails on 2 compilers ghs and iar	Impact/Effect: Build fails on IAR and GHS compilers due to a missing definition of the ISR() macro. This prevents successful compilation of modules that rely on interrupt service routines. Root Cause: The ISR() macro is not defined in the current context. This macro is typically provided by OS	3 - Moderate (apparent functional problem)

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Issue key	Summary	Notes for Integrator	Severity
		abstraction headers. Workaround: Include the appropriate OSIF header (Osif_Internal.h or Osif.h) in the affected source file to resolve the build error without modifying the macro manually.	
ARTD-114360 pwm	[PWM][ETPU] Output signal fails when set Period is greater than 16 bits and Duty is 100%	Impact/Effect: Incorrect casting may lead to runtime errors or unexpected behavior if values exceed the valid range of the target data type. Workaround: Ensure values used remain within the valid range of the current data type to avoid issues until the fix is applied.	3 - Moderate (apparent functional problem)
ARTD-114272 pwm	[PWM] Tresos generation errors	Impact/Effect: PWM functionality using ETPU IP is not supported in the P19 release. Therefore, the ETPU driver does not need to be added. Workaround: For P19, use PWM with supported IPs only. Do not include the ETPU driver.	3 - Moderate (apparent functional problem)
ARTD-114127 pwm	[PWM][PWM_ETPU] Missing resources on other derivative S32 K396_289 Pins	Impact/Effect: This issue was discovered during development and is isolated to this release. Certain resources are only available for the S32K396_MAPBGA289 derivative and are missing in other derivatives. Workaround: This issue was discovered during development and is isolated to this release, but in case is needed use only S32K396_MAPBGA289 derivative	3 - Moderate (apparent functional problem)
ARTD-105646 pwm	[PWM][PWM_ETPU] function "Eptu_Pwm_Ip_IrqHandler" has no prototype when build on IAR compiler	Impact/Effect: The customer may experience unexpected behavior or runtime errors due to incorrect assumptions about the function's parameters or return type, potentially leading to system instability or malfunction. Workaround: Declare the function prototype explicitly in a header or source file before its usage to ensure correct compilation and predictable behavior.	3 - Moderate (apparent functional problem)
ARTD-180522 CanTrcv	[CanTrcv] Implement check on partition assigned for correctness or out of bounds accesses	Implement ECUC partition Check to don't exceed no of partition configurated.	3 - Moderate (apparent functional problem)
ARTD-180600 lin	[lin] Implement check on partition assigned for correctness or out of bounds accesses	Implement ECUC partition Check to don't exceed no of partition configurated	3 - Moderate (apparent functional problem)

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Issue key	Summary	Notes for Integrator	Severity
ARTD-71747 mem_eep	[S32K358][MEM_EEP] Build fail with VARIANT_NO=3 on EB and CT	impact/effect: build failure when configuring postBuildVariant and variant > 1. workaround: The driver driver supports only 'VariantPre Compile' as ConfigurationVariant.	3 - Moderate (apparent functional problem)
ARTD-71410 mem_eep	[S32K3 3.0.0 P07][Mem_Eep] Remove redundant codes which support HS200 HS400 in S32K3XX platform	impact/effect: No impact. Just remove redundant codes which support HS200 HS400 in S32K3XX platform workaround: None	3 - Moderate (apparent functional problem)
ARTD-67362 mem_infls	[MEM_INFLS] Wrong VendorApi Infix in .xdm	impact/effect: The VendorApi Infix value in the interface can be disabled workaround: Alway enable VendorApiInfix node in interface	3 - Moderate (apparent functional problem)
ARTD-62493 memacc	[SAF85_S32R41 1.0.0][MemAcc] MemAcc_GetJobResult return not correct value when compare operation determined a mismatch	impact/effect: Adjust the return result for the MemAcc_GetJob Result function to return MEMACC_ MEM_INCONSISTENT instead of MEMACC_MEM_FAILED when the comparison operation identifies a mismatch. workaround: No workaround	3 - Moderate (apparent functional problem)
ARTD-61591 mem_exfls	[MEM_EXFLS] Bulk data read is not stable on K358, on HyperFlash board	Fix issues with Read from Hyperflash showing wrong results after a number of blocks.	3 - Moderate (apparent functional problem)
ARTD-200527 mem_eep	[Mem_Eep] Report DEV_ERROR_ DETECT is incorrect when calling FastBoot of Mem_43_EEP_Hw SpecificService	impact/effect: updated Mem_43_ EEP_HwSpecificService function call FastBoot don't report MEM_43_ EEP_E_UNINIT workaround: None	3 - Moderate (apparent functional problem)
ARTDCC2-290 spi	SPI Driver: tDT is added in CS Continuous Mode with CS_VIA_ GPIO	impact/effect: Modified the SPI/ DSPI configuration code to support continuous chip select behavior for GPIO CS mode Moved the CONT bit configuration before the chip select mode selection in all configuration files workaround: no impacted	3 - Moderate (apparent functional problem)
ARTDCMCU-540 messaging	[Messaging] RxTxSkipDefault Handler and GPISkipDefault Handler values should be consistent in some MU instances	impact/effect: EBT/S32CT does not raise an error when Messaging MuRxTxSkipDefaultHandler and MessagingMuGPISkipDefault Handler are configured with different values for RTUn MUB instances. This may leads to a build failure. workaround: Ensure that MessagingMuRxTxSkipDefault Handler and MessagingMuGPISkip DefaultHandler are configured with the same value for RTUn MUB instances.	3 - Moderate (apparent functional problem)
ARTDCC3-264 LIN	[S32K3][LIN] Remove the redundant code in IRQ handler function for IP FLEXIO	impact/effect: There are not any impact to customer workaround: None	3 - Moderate (apparent functional problem)

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Issue key	Summary	Notes for Integrator	Severity
ARTD-200424 sent	[S32K3 6.0.0 P02][SENT][K356] Name must be a valid identifier in IP SENT layer	impact/effect: Implemented by [ARTD-189605] workaround: Implemented by [ARTD-189605]	3 - Moderate (apparent functional problem)
ARTD-177162 spi	SPI endif positioned wrongly	fixed in BLN_RTD_4.7_S32K3XX_S32M27x_4.0.0_FBR_2.2	3 - Moderate (apparent functional problem)
ARTDCC2-73 spi	[spi] Implement check on partition assigned for correctness or out of bounds accesses	impact/effect: Driver now can cover the case when GetID API return value is not valid, then the issue about array index is out of bound will be eliminated workaround: no workaround needed	3 - Moderate (apparent functional problem)
ARTD-198334 mcu	[MCU] M4_SRC_LICENSE_INDICATOR not suited for usage in clock_utils.js	impact/effect: The header that is written by clock_utils.js will be moved directly into the generated files. workaround: NA	3 - Moderate (apparent functional problem)
ARTDCC3-121 LIN	[Lin] pointer assumed to be nonzero - test removed	impact/effect: <Add one sentence explaining how customer is impacted> workaround: <Add one sentence describing potential workaround>	3 - Moderate (apparent functional problem)
ARTDCC3-122 LIN	[Lin] overflow in constant expression of type "unsigned long int	impact/effect: <Add one sentence explaining how customer is impacted> workaround: <Add one sentence describing potential workaround>	3 - Moderate (apparent functional problem)
ARTDCMCU-429 MCU	[MCU]wrong variable mapping in Clock_Ip.c	impact/effect: Clock_Ip_pfk NotificationsCallback is allocated by default in data segment. workaround: Without modifying the source code a linker script could allocate Clock_Ip_pfkNotifications Callback to a specific memory section.	3 - Moderate (apparent functional problem)
ARTDCMCU-147 MCU	[MCU][K3] Clocks tool allow division value more than RM	impact/effect: Divisions greater than 8 are not supported by hardware; they will not function correctly if configured. workaround: not use Divisions greater than 8 with Mux 8 on interface	3 - Moderate (apparent functional problem)
ARTDCMCU-88 MCU	[MCU][S32K348] Clocks tool interface issue	impact/effect: wrong clock MUA_CLK and MUB_CLK in clock tool S32K3 workaround: use MCU HLD for both clocks	3 - Moderate (apparent functional problem)
BENV-604 base	[BASE] BaseNXP.h file contains wrong header file include name	impact/effect: The case-sensitive issue is fixed so the compliance issue is resolved. This file Base NXP.h initially be required by Elektrobit and it seems that only EB needs this file. RTD package does not use this file so there is nothing	3 - Moderate (apparent functional problem)

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Issue key	Summary	Notes for Integrator	Severity
		impact to functionality. workaround: N/A	
SRTDCCOM-834 uart	[S32K3][UART] Remove the redundant code in IRQ handler function for IP FLEXIO	impact/effect: There are not any impact to customer workaround: None	3 - Moderate (apparent functional problem)
SRTDCCOM-906 i2s	[I2S] Build fail because the driver i2s generates redundant file	impact/effect: Exclude all Mcl include files when Mcl is not a dependency module workaround: Add Mcl to project	3 - Moderate (apparent functional problem)
SRTDCCOM-1005 sent	[S32K5 0.8.0 EAR][SENT] Too many initializer values for Flexio_ Sent_Ip_apxBase	impact/effect: Build fail when the SentHwControllerRef node is only one in the configuration workaround: Add 2 the SentHw ControllerRef node to avoid build fail	3 - Moderate (apparent functional problem)
SRTDCCOM-701 i2s	[I2S][SAI] Can not send polling with SAI	impact/effect: Adding POLLING mode for Sai transfer type, with Blocking Sending Transfer workaround: N/A	3 - Moderate (apparent functional problem)
SRTDCCOM-1103 i2s	[S32K396] S32DS configuration issue in SPI module when I2S module is present	impact/effect: User can not generate when adding SPI and I2s in the same project. All derivatives that have FlexIO and do not have SAI are impacted. workaround: user should use SPI and I2s separately on DS configuration.	3 - Moderate (apparent functional problem)
ARTDCMEM-458 Mem_ExFls	[Mem_Exfls] Generate failed when MemAcc config multi group on DS	impact/effect: Generate fail when configuring 2 or more Qspi_Ip/ Xspi_Ip in 2 or more functional group on DS workaround: Configure only one Qspi/Xspi component in one functional group	3 - Moderate (apparent functional problem)
ARTDCMCU-1087 MCU	[MCU]CMU_FC configuration options are not aligned with S32K3RM	impact/effect: Unable to configure CMU_FC destructive reset to occur. workaround: Use interrupts instead.	3 - Moderate (apparent functional problem)
ARTDCMEM-490 Mem_InFls	[mem_infls] Implement check on partition assigned for correctness or out of bounds accesses	Update missing Det error reporting in case Driver Service is called from a different Core instead of the one that initialized the driver.	3 - Moderate (apparent functional problem)
ARTDCMEM-560 Fee	[Fee] Add driver initialize check for the functions Fee_GetBlockData Status	impact/effect: Fee_GetBlockData Status will return correct status when the driver is not initialized. workaround: No	3 - Moderate (apparent functional problem)
ARTDCMEM-684 MemAcc	[S32K5_0.8.0][EAR] No error appears when the Prefix name is incorrectly entered on S32DS	impact/effect: Build can be failed when config Prefix name is incorrectly workaround: can not workaround, user need to config Prefix name is correctly	3 - Moderate (apparent functional problem)
ARTDCMEM-454 Mem_Eep	[Mem_Eep] eMMC driver initialization issue	impact/effect: In some cases, initializing eMMC in DDR52, HS200 or HS400 mode may have a	3 - Moderate (apparent functional problem)

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Issue key	Summary	Notes for Integrator	Severity
		certain failure rate. This is because sending CMD13 during the timing change process (CMD6) is not yet synchronized. workaround: If this issue occurs, a temporary workaround is to operate at a lower clock frequency.	
ARTDCMEM-456 MemAcc	[MemAcc] build failed when config multi variants on both DS,EBT	impact/effect: build will be failed when user config 2 variant(vs0: use aboth Mem_Exfls, Mem_Eep to config MemAccAddressArea, Vs1: only use Mem_Eep to config Mem AccAddressArea workaround: can not workaround	3 - Moderate (apparent functional problem)
ARTDCMEM-391 Mem_InFls	[S32K3XX_M27X 7.0.0][MEM_INFLS][S32K389] Wrong CRC calculation	Revise the CRC calculation function: Avoid hardcoding values like this: [!VAR "MemUnitIndex" = "0"!][!//] Instead, the value should be retrieved dynamically from the actual node.	3 - Moderate (apparent functional problem)
ARTDCC2-93 i2c	[LPI2C] Glitch filter remains enabled when it is disabled.	impact/effect: For enabling glitch filters on master mode, the checkbox must be checked. Glitch filters for SDA/SCL lines equals 0 if glitch filter is not enabled workaround: Number of cycles is not reverted to zero in glitch filter even if it is disabled. The values must be input manually with 0 and then disable it.	3 - Moderate (apparent functional problem)
ARTDCC1-168 eth	[ETH][GMAC] Wrong data read because Driver invalidates Cache memory for only the first descriptor in the chain when S/G enabled	impact/effect: Rx queue will not work as expected when Scatter Gather and Cache Management features enable at same time workaround: None	3 - Moderate (apparent functional problem)
ARTDCT2-760 gpt	[GPT] Issue with PIT max value of LDVALn[TSV] register	impact/effect: The Pit_Ip_Start Channel function needs to check the value of the countValue variable (it must be greater than 0). Passing the value 0 to the LDVAL register may cause an error. workaround: Pass a countValue greater than 0 to the Pit_Ip_StartChannel function.	3 - Moderate (apparent functional problem)
ARTDCC1-236 eth	Buffer descriptors not correctly released when using multi-buffer send	impact/effect: Eth_43_GMAC_SendMultiBufferFrame might not working if requested to transmit the same buffer or Eth_43_GMAC_TxConfirmation is called when the buffers are only partially transferred, not completely. Eth_43_GMAC_SendMultiBufferFrame and and Eth_43_GMAC_Provide TxBuffer/Eth_43_GMAC_Transmit should not be used in the same	3 - Moderate (apparent functional problem)

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Issue key	Summary	Notes for Integrator	Severity
		time workaround: The user should request to transmit different buffers when using multiple transmit feature. Eth_43_GMAC_Tx Confirmation should be called when all buffers completed transfer or using INTERRUPT mode. Eth_43_GMAC_SendMultiBufferFrame and and Eth_43_GMAC_Provide TxBuffer/Eth_43_GMAC_Transmit should not be used in the same time	
ARTD-204795 rte	[CESS-249] File names SchM_Motor_control.[hC] correction	impact/effect: case sensitive issue so cannot work correctly in Linux environment workaround: update case sensitive for SchM Motor_Control	3 - Moderate (apparent functional problem)
ARTDCC1-256 eth	[ETH] Check the interrupt handler function in context of new R23-11 ASR version	impact/effect: - The Callback function of ISR was reverted - Eth_Cfg.c was created to store the generated IngressQueueHandler Function which is configured in Eth CtrlConfigIngressQueueHandler Function by the user workaround: N/A	3 - Moderate (apparent functional problem)
ARTD-205005 can	[CAN]Missing case CAN_RX_ENHANCED_FIFO in function Can_lpw_SetControllerToStop Mode	impact/effect: This may lead to incorrect behavior when the controller is configured with enhanced RX FIFO mode. Solution: Update the function Can_lpw_SetControllerToStopMode to handle the missing case CAN_RX_ENHANCED_FIFO	3 - Moderate (apparent functional problem)
ARTD-205132 can	[CAN]Missing case CAN_RX_ENHANCED_FIFO in function Can_lpw_SetControllerToStop Mode	impact/effect: Can hard faulted workaround: update the function *Can_lpw_SetControllerToStop Mode* to handle the missing case CAN_RX_ENHANCED_FIFO	3 - Moderate (apparent functional problem)
ARTDCMCR-110 mcl	[MCL][S32K3] S32CT generate unexpected code after importing epc file	impact/effect: <Add one sentence explaining how customer is impacted> workaround: <Add one sentence describing potential workaround>	3 - Moderate (apparent functional problem)
ARTDCC3-245 CAN	[CAN] Implement ERR052558	impact/effect: Message buffer (MB) overrun status is cleared when reading Enhanced RX FIFO (ERF) workaround: - Update in Can_Hw ObjectConfigType: uint8 Can_u8 HwBufferIndex to uint8 * Can_p HwBufferIndex - Update driver to align with the the change above - Update and optimize in generated code to generate exactly the buffer address and buffer index which are	3 - Moderate (apparent functional problem)

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Issue key	Summary	Notes for Integrator	Severity
		mapped to hw object configuration: - generate code for HLD: validate hw object configuration, check and generate MB index and MB address of each hw object (only for legacy fifo and normal MB) - generate code for IPL: calculate the exact number of MB to be used - Add restricted MB list in resource which are not be used when enabling enhanced fifo to avoid overrun status is cleared unexpectedly following ERR052558	
ARTDCANG-356 adc	[S32K3 RTD 4.7 5.0.0] [ADC] Group conversion doesn't finish if add ADCMuxDelay	impact/effect: Customer not impacted. workaround: For K34x device, either set Adc Mux Delay to 0 or use a prescale value of 1 or 2. Updated the configurator warnings for K34x devices.	3 - Moderate (apparent functional problem)
ARTD-72656 gpt	[GPT][PIT][WDOG sporadic reset] Wdg doesn't receive the interrupt from GPT	the handling of spurious interrupts at PIT IPL was updated; missing exclusive areas in IPL functions were added	3 - Moderate (apparent functional problem)
ARTD-72646 gpt	[GPT][NXPIS-194][RTC] Investigate issue related to RTC_IP_TICKS_PER_SECOND is always 0 on EB project	corrected generation for RTC_IP_TICKS_PER_SECOND value in EB project	3 - Moderate (apparent functional problem)
ARTD-71221 gpt	[GPT][NXPIS-215][GetTime Elapsed] Issue related to static variable declaration inside Gpt_Get TimeElapsed()	removed static from var returnHw ChannelInfo which is used in Gpt_GetTimeElapsed()	3 - Moderate (apparent functional problem)
ARTDCT2-509 gpt	[GPT] Fix some issue from SA finding - Gpt code	impact/effect: Regarding PIT ISR's, there are unused functions that increase source code size, not all being used. Some critical sections are entered without needing to. workaround: does not apply as this is just an improvement	3 - Moderate (apparent functional problem)
ARTDCT2-507 gpt	[GPT] STM timer Call back not triggered if resolution is less than 6 micro sec	impact/effect: for timer instance STM_0 looks like interrupt is not triggered in case of timer resolution is configured to less then 6 micro seconds workaround: Start STM timer earlier than 6 us before counter value would reach interrupt threshold	3 - Moderate (apparent functional problem)
ARTDCMCU-1278 MCU	S32K344 LQFP 100 package, clock settings is not working	impact/effect: When adding the MCU component to an S32K344 LQFP100 package S32DS project, then changing the clock source for a mux in Clocks tool, an unresolvable error will appear in the MCU component. workaround: applicable for project has 1	3 - Moderate (apparent functional problem)

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Issue key	Summary	Notes for Integrator	Severity
		Clock configuration in Clocks tool. Solution1: Remove Mcu component. Do the configure all clock settings in Clocks tool, then add the Mcu component back Solution2: In McuClockSetting Config tab, add 1 more element by pressing "+" button, in new Mcu ClockSettingConfig change the name of new configuration e.g. from McuClockSettingConfig_1 to McuClockSettingConfig_0. Then select McuClockSettingConfig index 0 and press "x" button to remove error element McuClock SettingConfig which contains error.	
ARTDCMCU-1266 MCU	[MCU]not raise error when value of some node different for many clock config	impact/effect: Error not raised when the value of the node is different between clock configurations. workaround: Check all functional group in S32DS for general items	3 - Moderate (apparent functional problem)
ARTDCMCU-1067 MCU	[Mcu][S32K3]The MC_ME slot display is missing the gate register element for the AES clock	impact/effect: The MC_ME slot display is missing the gate register element for the AES clock. It does not impact enable /disable gate functionality. Not showing the correct value in interface. workaround: Cosmetic issue	3 - Moderate (apparent functional problem)
ARTDCMCU-617 MCU	[MCU][S32K3]improve function Clock_Ip_SetCmuFcFceRefCnt LfrefHfref	impact/effect: No impact, ticket raised to improve the the function Clock_Ip_SetCmuFcFceRefCnt LfrefHfref workaround: No need for a workaround as there is no issue.	3 - Moderate (apparent functional problem)
ARTDCMCU-517 MCU	[MCU] Clock_Ip_CallbackPllEmpty Disable is redundant	impact/effect: No impact, in order to improve CCOV unused functions need to be removed. workaround: No need for a workaround as there is no functional issue	3 - Moderate (apparent functional problem)
ARTDCMCU-193 MCU	[MCU][S32K3-S32M27x] wrong implement clock source CGM_ MUX11 in interface	impact/effect: On K358, when writing 0x1900_0004 to mux7 CSC it will cause a bus error; 0x19 is corresponding to GMAC_ MII_RGMII_RX_CLK source. workaround: Do not use this clock source.	3 - Moderate (apparent functional problem)
ARTDCMCU-187 MCU	[MCU] Correct implementation for ERR050583	impact/effect: driver does not implement correctly ERR050583 (about K394 and k364), not fully implement (about clear fes before write to ferd) workaround: There is no workaround for this issue.	3 - Moderate (apparent functional problem)
ARTDCMCU-78	[MCU] Clarify information pll clock range S32K3xx	impact/effect:user can not config for some clock mode option : 1, S32K358 mode B, F: Invalid	3 - Moderate (apparent functional problem)

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Issue key	Summary	Notes for Integrator	Severity
MCU		division value for QSPI_SFCK_CLK QSPI_SFCK_CLK is sourced from PLL_PHI1_CLK (MC_CGM.MUX_10_CSC[SELCTL] and MC_CGM.MUX_11_CSC[SELCTL] must equal 1001b.) But when PLL_PHI1_CLK = 240Mhz/160 Mhz, QSPI_SFCK_CLK can't be 120Mhz/80Mhz 2, divider range for TRACE_CLK between clock diagram and clock option (ex S32K388 range of division value is 1 to 8 but in clock option can div 10) workaround: for issue 2: go to mcu.xdm and remove/changed condition divider range for TRACE_CLK	
ARTDCMCU-60 MCU	[MCU]Mcu_GetPllStatus still checking PLLAUX when it not under mcu control	impact/effect: Mcu_GetPllStatus is checking PLLAUX when it not under mcu control workaround: No workaround available	3 - Moderate (apparent functional problem)
ARTDCMCR-605 mcl	[MCL] S32DS generate failure after added Mcl component	impact/effect: The customer is unable to use S32DS to configure the drivers workaround: n.a. / if possible use e.g. EB Tresos	3 - Moderate (apparent functional problem)
ARTDCMCR-1005 crc	[crc] [S32K3]Wrong mask in function Crc_Ip_TableCalculate CRC32_Slice_By_8	impact/effect: cannot use slice by 8 algorithm with table mode workaround: NA	3 - Moderate (apparent functional problem)
ARTDCMCR-413 mcl	[s32k5][mcl][emios] build fail when using interrupt function	The EMIOS issue has been resolved. The EMIOS of the MCL driver is working without any errors.	3 - Moderate (apparent functional problem)
ARTDCMCU-250 MCU	[MCU] Sync partition ID on Clock Tool and MCU component	impact/effect: The partition ID on Clock Tool and the MCU component must be the same, but the interface allows different values. workaround: need to update the partition ID in Clock Tool and the MCU component so they are the same.	3 - Moderate (apparent functional problem)
ARTDCANG-1522 dio	[S32K3][DIO] Partition ID is cast to uint8 which can lead to truncation with >255 partition	impact/effect: Potential issues when using >255 partitions. workaround: Use 255 or less partitions.	3 - Moderate (apparent functional problem)
ARTDCMCU-249 MCU	[MCU]: Core power down when call Mcu_Init_Clock with VS_2	impact/effect:core power down may occur when call Mcu_InitClock with VS_2 for S32K396,S32K344,S32K358 and S32K388 workaround: issue still investigate.not have any workaround	3 - Moderate (apparent functional problem)
ARTDCMCU-45 MCU	[Clock][K3] CLOCK_IP_DIVIDER_TRIGGERS_COUNT has different values between eb and ct	impact/effect: no impact. workaround: NA	3 - Moderate (apparent functional problem)

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Issue key	Summary	Notes for Integrator	Severity
ARTDCANG-1042 port	[PORT] FCCU EOUT signal issue	impact/effect: Change sequence in Siul2_Port_Ip_PinInit to wire MSCR first before wirite IMCR workaround: No workaround required	3 - Moderate (apparent functional problem)
ARTDCANG-1040 port	[PORT] S32K312 CMP0 inputs mismatch	impact/effect: Can not set CMP0_IN4 mode is supported for pin PTC3. workaround: Set manually by Siul2 register address, such as SSS/OBE/IBE bit on PTC3	3 - Moderate (apparent functional problem)
ARTDCT2-759 gpt	[GPT]RTI has not completed configuration yet before going into Standby mode	impact/effect: After configuring PIT_RTI, you need to wait and check if PIT_RTI is actually working before you can enter standby mode. Entering standby mode while the RTI is not ready and without checking the RTI status may result in the RTI not working after waking up. workaround: N/A; please use Pit_StartChannel instead of Pit_EnableTimer as StartChannel has already implemented in code the wait for PIT_RTI.	3 - Moderate (apparent functional problem)
ARTD-180057 wdg	[WDG] endif positioned wrongly	impact/effect: Wdg_EnvCfg cannot be used in C++ mode workaround: Do not use driver with c++ or manually add at the end of Wdg_EnvCfg.h the following lines of code	3 - Moderate (apparent functional problem)
ARTDCC2-91 i2c	[I2C] S32K3 LPI2C slave bus hang issue	impact/effect: When using s32K as LPI2C target with another device as a controller and multiple targets on the bus, due to multiple access, S32K I2C target can remain in stuck low. workaround: No workaround	3 - Moderate (apparent functional problem)
ARTDCMCU-854 messaging	[Messaging][S32N_2.0.0] Build failed when configuring skip handler	impact/effect: Implicit declaration of function 'Mu_Messaging_Ip_RxSpuriousInt_Handling' when IP_RTU0_DSCMU0_MUB instance is configured with skip handler. workaround: Do not skip handler in configuration.	3 - Moderate (apparent functional problem)
SRTDCCDD-2034 ae	[AE][S32K3 6.0.0 QLP05] E_PARAM_CONFIG is not reported	impact/effect: When the partition ID does not match the configuration or exceeds AE_MAX_PARTITIONS, the E_PARAM_CONFIG DET error is not reported except in the Ae_Init() function. workaround: Always call the Ae_Init() function first before any other function, or ensure the partition matches the configuration before calling a function.	3 - Moderate (apparent functional problem)

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Issue key	Summary	Notes for Integrator	Severity
SRTDCCDD-2279 ae	[AE][S32K3 7.0.1] E_PARAM_CONFIG is not reported	impact/effect: When the partition ID does not match the configuration or exceeds AE_MAX_PARTITIONS, the E_PARAM_CONFIG DET error is not reported except in the Ae_Init() function. workaround: Always call the Ae_Init() function first before any other function, or ensure the partition matches the configuration before calling a function.	3 - Moderate (apparent functional problem)
ARTDCANG-631 dio	[DIO][Except K3] Commented MISRA violations that need to be fixed	impact/effect: Guarded macros in order to fix static analysis violations. workaround: No workaround necessary.	3 - Moderate (apparent functional problem)
ARTDCANG-1433 port	[PORT] Cannot configure 2 pins with ONLY_INPUT mode	impact/effect: when pin modes are changed at runtime between two input modes the previous IMCR register is not cleaned before the new IMCR register is set therefore the pin will be configured with both input modes workaround: N/A	3 - Moderate (apparent functional problem)
BENV-590 base	[S32K396] USE_$(MODULE)_MODULE is set to STD_ON when module is disabled on S32DS	impact/effect: <Add one sentence explaining how customer is impacted> workaround: <Add one sentence describing potential workaround>	3 - Moderate (apparent functional problem)
ARTDCMCR-855 mcl	[MCL][S32N] DMA ISR function does not clear the interrupt flag when driver is not initialized and multi partition is enabled	When Multipartition support is enabled, the ISR function does not clear the DMA interrupt flag in the case where the driver has not been initialized. So user need to manage the DMA interrupt flag in this case by themselves. There is no workaround in the driver.	3 - Moderate (apparent functional problem)
ARTDCMCR-113 crc	[crc] Implement check on partition assigned for correctness or out of bounds accesses	impact/effect: may cause user's code to behave in unexpected ways workaround: Users should ensure to verify the partition ID during configuration.	3 - Moderate (apparent functional problem)
ARTD-1712 wdg	[WDG] Lock configuration when SetMode = OFF has no effect	Issue was fixed in https://bitbucket.sw.nxp.com/projects/ARTD/repos/wdg/pull-requests/25/overview.	4 - Minor (cosmetic or unapparent problem)
ARTD-107815 fee	[fee] Driver makefile needs to be updated with the correct version of AUTOSAR SWS specification	impact/effect: only Fee.mak file (Updated driver with the correct version of Autosar (4.4)) so no impact workaround: cáº¡n not workaround	4 - Minor (cosmetic or unapparent problem)
ARTD-61632 i2s	[S32K3 3.0.0][S32K1 2.0.0] i2s: Node "FlexioI2sCallback" can not generate ecvd in IP layer FLEXIO_I2s	ECVD generation is fixed. Options field is added to existing node.	4 - Minor (cosmetic or unapparent problem)

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ARTD-192956 mcl	[MCL] Error on ecvd file	impact/effect: Update default short name of IPL component to the same with IPL component name workaround: No impact	4 - Minor (cosmetic or unapparent problem)
ARTDCC2-63 spi	[spi] dead assignment to "<variable>" eliminated	impact/effect: no impacted to user, just refactor to remove dead code workaround: no workaround needed	4 - Minor (cosmetic or unapparent problem)
ARTDCMCR-119 crc	[crc] dead assignment to "<variable>" eliminated	impact/effect: not impact to user workaround: Variables are used to increase code transparency and should not be removed.	4 - Minor (cosmetic or unapparent problem)
SRTDCCOM-822 i2s	[I2s] Update macro error when UserID is greater than the maximum number of partitions	impact/effect: Bug does not affect the user. It affects only Det errors. workaround: No workaround needed.	4 - Minor (cosmetic or unapparent problem)
ARTDCMCR-127 mcl	[mcl] dead assignment to "<variable>" eliminated	impact/effect: n.a workaround: n.a	4 - Minor (cosmetic or unapparent problem)
ARTDCC2-101 i2c	[i2c] dead assignment to "<variable>" eliminated	impact/effect: remove compiler warning when dev error detection is disabled workaround: N/A	4 - Minor (cosmetic or unapparent problem)
SRTDCCOM-824 sent	[Sent] Update macro error when UserID is greater than the maximum number of partitions	impact/effect: Bug does not affect the user. It affects only Det errors. workaround: No workaround needed.	4 - Minor (cosmetic or unapparent problem)
SRTDCCOM-843 sent	[SENT] Check the details of requirement CPR_RTD_00012.sent to clean up the code/design/FMEA	public APIs : Update the design to include the note: "This interrupt is handled by the MCL driver" not by the SENT driver in the Flexio_Sent_Ip_IRQHandler function. Configuration ASR/nonASR : NA Notes for integrators : NA	4 - Minor (cosmetic or unapparent problem)
ARTDCMCU-1366 MCU	[MCU] Update node info form MCU component with the allowed attributes	impact/effect: No impact, improvement by removing unnecessary attributes from peripheral configuration. workaround: No need for a workaround as this is not an issue.	4 - Minor (cosmetic or unapparent problem)

4 Changes List for S32K3 RTD 7.0.1

4.1 Change List for 7.0.1

Issue key	Issue Type	Summary	Notes for Integrator
SRTDCCOM-2024 sent	Task	[S32K3 7.0.1] [SENT] Add resources for s32k314_lqfp100 and s32k324_lqfp100	public APIs : No Configuration ASR/nonASR : No Notes for integrators : Update resource
SRTDCCOM-2022 uart	Task	[S32K3 7.0.1] [UART] Add resources for s32k314_	public APIs : No Configuration ASR/nonASR : No Notes for integrators : Update resource

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Issue key	Issue Type	Summary	Notes for Integrator
		lqfp100 and s32k324_ lqfp100	
SRTDCCDD-1929 ocotp	Task	[S32K3][ocotp] Add support for K314 LQFP 100 and K324 100 LQFP phantom devices	public APIs : The public APIs haven't changed. The current functionality is preserved. Configuration ASR/nonASR : Configuration unchanged. The current functionality is preserved. Notes for integrators : No additional information required. New resource files added for the S32K314 and S32K324. No additional changes.
SRTDCCDD-1421 ocotp	Task	[S32K3 7.0.0 QLP04][ocotp] Driver examples	public APIs : The public APIs haven't changed. The current functionality is preserved. Configuration ASR/nonASR : Configuration unchanged. The current functionality is preserved. Notes for integrators : No additional information required. The examples configuration and source code were updated to match the latest driver implementation.
SRTDCCDD-1418 gdu	Task	[S32K3 7.0.0 QLP04][gdu] Driver examples	public APIs : The public APIs haven't changed. The current functionality is preserved. Configuration ASR/nonASR : Configuration unchanged. The current functionality is preserved. Notes for integrators : No additional information required. The example configuration and source code were updated to match the latest implementation of the driver.
SRTDCCDD-1415 dpga	Task	[S32K3 7.0.0 QLP04][dpga] Driver examples	public APIs : The public APIs haven't changed. The current functionality is preserved. Configuration ASR/nonASR : Configuration unchanged. The current functionality is preserved. Notes for integrators : No additional information required. The example configuration was updated to use the latest driver settings in the project.
SRTDCCDD-1412 ae	Task	[S32K3 7.0.0 QLP04][ae] Driver examples	public APIs : The public APIs haven't changed. The current functionality is preserved. Configuration ASR/nonASR : Configuration unchanged. The current functionality is preserved. Notes for integrators : No additional information required. The configurations of examples were updated with the latest project settings.
SRTDCCDD-1376 ae	Task	[S32K3 7.0.0 QLP04][ae] Generate and fix quality documents(Dev) for the module	public APIs : The public APIs haven't changed. The current functionality is preserved. Configuration ASR/nonASR : Configuration unchanged. The current functionality is preserved. Notes for integrators : No additional information required. The User and Integration manuals were updated to reflect the latest changes in the driver.
BENV-843 base	Task	[Base][S32K3 7.0.1] Add support new sub-derivatives	public APIs : No APIs are added/removed or changed. Configuration ASR/nonASR : No configuration nodes are changed/added/removed. Notes for integrators : S32DS support is available for new sub-derivatives.
BENV-832 resource	Task	[S32K3 7.0.1][RESOURCE] Generate and fix quality documents(dev) for the module	public APIs : No APIs are added/removed or changed. Configuration ASR/nonASR : No configuration nodes are changed/added/removed. Notes for integrators : This ticket covers updating the UM/IM documentation for the S32K3 7.0.1 release.
ARTDCT2-764 gpt	Bug	[gpt] Issue in the implementation of MR200.	public APIs : no changed Configuration ASR/nonASR : no changed Notes for integrators : Emios_GPT can be configured at the same time with PWM and OCU
ARTDCT2-499 gpt	Bug	[Gpt] overflow in constant expression of type "unsigned long int	'- Rebuild test with TimeElapsed in compilers to double check wanning
ARTDCT1-720 icu	Bug	[icu]Generation failed on the interface when using the Emios_Icu and Icu component	Public APIs: No changes to existing APIs. Configuration (ASR/non-ASR): No changes in configuration. Notes for Integrators: No impact on integration.
ARTDCT1-478 pwm	Bug	[pwm] EMIOS_PWM_IP_ BUS_BCDE no longer exists in RTD 7.0.0 , generates unsolvable error in S32CT	Public APIs: N.A. Configuration ASR/nonASR : N.A. Notes for integrators: Added support for both _BCD and _BCDE masterbus types in the expressions used to check if Emios configuration is valid.
ARTDCT1-54 icu	Bug	[icu] Unserviced timebase interrupt	public APIs : No changes to existing APIs. Configuration ASR/nonASR : Added the Icu OverflowNotificationApi node to the Emios_Icu component and updated the corresponding macro definition Notes for integrators: The IcuOverflowNotificationApi node can be configured at the low level, this node is enabled by default.
ARTDCT1-49 pwm	Bug	[pwm] Pwm_Init generates a spike in the output signal.	public APIs : No impact Configuration ASR/nonASR : No impact Notes for integrators: From 2nd variant, using GPIO mode to set the logic in duty cycle: 0%, 100% when multi-variants are used
ARTDCMEM-1042 Fee	Story	[FEE] Create Driver Examples for S32K3 7.0.0 QLP04	public APIs: Nothing change Configuration ASR/nonASR: Nothing change Notes for integrators: Just add support examples
ARTDCMEM-821 Mem_InFls	Bug	[Mem_Infls] [S32K3 RTD 6.0.0] Mem_Infls C40_Ip_ AccessCode causing hard-fault	public APIs: No APIs are added/removed (high or low level) Configuration ASR/nonASR: No configuration nodes changed/added/removed Notes for integrators: No impact of changes for integrations and propose migration steps
ARTDCMEM-612 MemAcc	Bug	[MemAcc] Add note in the UM about the absolute address of Mem Services	public APIs: Nothing changes Configuration ASR/nonASR: Nothing changes Notes for integrators: Just add some update on UM relates to Dynamic Mem driver and Mem driver binary image format chapter. Address of function pointers must be absolutely address on RAM
ARTDCMEM-200 Mem_Eep	Story	[Mem_Eep] SW components with independent resources for RTD ART	public APIs: No change Configuration No change Notes for integrators: resource will be moved to driver plugin instead of resource plugin
ARTDCMEM-31 Mem_InFls	Story	[Mem_InFls] Support extra 64 KB Data memory on K388 derivative	new parameter added in order to extend memory for data flash on s32k388
ARTDCMCU-1062 MCU	Bug	[MCU]Generation failed on S32K389 package	public APIs : No change Configuration ASR/nonASR : No impact Notes for integrators : No impact

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Issue key	Issue Type	Summary	Notes for Integrator
ARTDCMCU-682 MCU	Bug	[Mcu] Crypto Ace triggering hardfault when clocks configured using MCU	public APIs : no change Configuration ASR/nonASR : update the MC_ME_slot on Mcu Peripheral PRTN1_COFB3_REQ112 replace to PRTN1_COFB3_REQ112_115 PRTN1_COFB3_REQ116 replace to PRTN1_COFB3_REQ116_119 PRTN1_COFB3_REQ120 replace to PRTN1_COFB3_REQ120_123 PRTN1_COFB3_REQ124 replace to PRTN1_COFB3_REQ124_127 PRTN2_COFB2_REQ72 replace to PRTN2_COFB2_REQ72_75 PRTN2_COFB2_REQ76 replace to PRTN2_COFB2_REQ76_79 PRTN2_COFB2_REQ80 replace to PRTN2_COFB2_REQ80_83 PRTN2_COFB2_REQ84 replace to PRTN2_COFB2_REQ84_87 PRTN2_COFB2_REQ88 replace to PRTN2_COFB2_REQ88_91 Notes for integrators : update MC_ME slot and update the code generate
ARTDCMCU-327 MCU	Bug	[MCU] Wrong clock assignment/ description in Clock configuration tool	public APIs : no change Configuration ASR/nonASR : update in Clock tool, No impact : - Change clock source for ADC_CLK to CORE_CLK as default value of ADCBIST , correct get clock frequency -remove TSENSE0_CLK and move clock source of TEMPSENSE_CLK to AIPS_SLOW_CLK , correct get clock frequency - change clock source for COOL_FLUX_x_RAM to CM7_CORE_CLK , correct get clock frequency -change clock source for ERM0 and ERM1 CLK to AIPS_PLAT_CLK , correct get clock frequency -change clock source LPSPI1/2/3/4/5 on K389 to AIPS_PLAT_CLK , correct get clock frequency - correct function get clock frequency for LPUART1 Notes for integrators : NA
ARTDCMCU-239 MCU	Bug	[MCU] Wrong implementation of the function Power_Ip_MC_ME_SocStandbyEntry	public APIs : No updated Configuration ASR/nonASR : No impact Notes for integrators : No impact
ARTDCMCR-983 crc	Story	[CRC] BSD 2-Clause license text must be included into the header of the Crc_Ip_Lookup_Tables_Slice_By_N.h	public APIs : no change Configuration ASR/nonASR : no change Notes for integrators : update the license text according to description
ARTDCMCR-864 rm	Story	[S32K3] [RM] Add resources for s32k314_lqfp100 and s32k324_lqfp100	Resource files for derivative s32k314_lqfp100 and s32k324_lqfp100 were added
ARTDCMCR-647 rm	Bug	S32K396: XRDC: SIUL virtual wrapper PDAC slots are not in line with the RM	Update resource file for XRDC to be in sync with RM.
ARTDCMCR-615 rm	Bug	[RM] [S32K3] Update driver resources to be in sync with RM (XRDC MRCs)	'- Updated number of descriptors, slot name and slot start/end address for XRDC MRC5 - Users need to update XRDC memory configuration, change Xrdc_0_PFC1_PFLASH0 and Xrdc_0_PFC1_PFLASH1 to Xrdc_0_PFC1_PFLASH.
ARTDCC2-807 i2c	Bug	[S32K3XX][I2C][FlexIO] Infinite loop in tx_handler on NAK	public APIs : No Configuration ASR/nonASR : No Notes for integrators: No
ARTDCC1-597 eth	Story	[S32K3 7.0.1]Prepare driver for S32K3 7.0.1 RFP	Updated examples Added resources for the new phantom sub-derivatives
ARTDCC1-49 eth	Story	[ENABLER] Immediate Transfer for GMAC platforms	public APIs : added support new API: Eth_ImmediateTransmit Configuration ASR/non ASR : changed to READONLY for un-supported nodes: EthCtrEgressHardwareSupported DataTransferThreshold EthCtrlEnableEgressHardwareSupportedDataTransfer Notes for integrators : please refer to Eth_ImmediateTransmit prerequisites chaper in UM
ARTDCANG-1185 dio	Story	[S32K3 7.0.1][DIO] Support new sub-derivatives K314 LQFP 100 and K324 100 LQFP	public APIs : Support new sub-derivatives K314 LQFP 100 and K324 100 LQFP Configuration ASR/nonASR : Support new sub-derivatives K314 LQFP 100 and K324 100 LQFP Notes for integrators : Support new sub-derivatives K314 LQFP 100 and K324 100 LQFP
ARTDCANG-1184 port	Story	[S32K3 7.0.1][PORT] Support new sub-derivatives K314 LQFP 100 and K324 100 LQFP	public APIs : Support new sub-derivatives K314 LQFP 100 and K324 100 LQFP Configuration ASR/nonASR : Support new sub-derivatives K314 LQFP 100 and K324 100 LQFP Notes for integrators : Support new sub-derivatives K314 LQFP 100 and K324 100 LQFP
ARTDCANG-1142 adc	Bug	[S32K341][ADC] AdcRead Group() return E_NOT_OK response	public APIs : N/A Configuration ASR/nonASR : N/A Notes for integrators : N/A By adding critical section protection when reading data from CDR, the issue was solved
ARTDCANG-953 adc	Bug	[ADC] SDADC configured with HW Trigger fails to generate Adc_Ipw_PBcfg.c	' public APIs : None changed Configuration ASR/nonASR : None changed Notes for integrators : No impact
ARTDCANG-474 adc	Bug	[S32K3XX][ADC][S32 K36x]The array ADC_IPW_ADC_TO_CTU_MAPPING_ARRAY was calculated incorrectly	public APIs : no API updated Configuration ASR/nonASR : no configuration update Notes for integrators : fixed code generation, no integration changes needed.
ARTDCANG-421 port	Bug	[port][S32K3] dead assignment to "<variable>" eliminated	public APIs : nothing impact Configuration ASR/nonASR : nothing impact Notes for integrators : fix some dead asssignment for tasking compiler
ARTDCANG-327 port	Bug	[PORT] ADC_0/1/2 not available in S32K36x	ADC_0/1/2 have been removed on K36X

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